Computational resources

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Recap

- Logging
  - MLflow
- Visualization
- Frameworks for training
  - Lightning
  - Hugging face

Outline

- Contours of computation
  - Offline
  - Online (realtime)
  - Near real time
  - Edge
- Types of compute resources
- Jobs runners

Contours of computation

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Offline

Operations:

- can take arbitrary time (from 100s of ms to 10s of days)
- can take arbitrary compute (from 1 core to 10k)
- can go down without business impact
- have a relatively low execution priority

Examples:

- Building daily analytics
- EDA
- Model training

Online (aka realtime)

Real-time computing (RTC) is the computer science term for hardware and software systems subject to a "real-time constraint", for example from event to system response. Real-time programs must guarantee response within specified time constraints, often referred to as "deadlines".

Online (aka realtime)

Operations:

- must meet estimated time of arrival (ETA) (usually < 1 second)
- use predefined amount of resources
- failure causes apps to break
- have highest priority

Examples:

- site rendering
- quotes API of a stock market
- ML models inference
    - search engine results
    - recommender systems
    - advertisements
    - route generation on maps
    - chat bots

Near real time

The term "near real-time" or "nearly real-time" (NRT), in telecommunications and computing, refers to the time delay introduced, by automated data processing or network transmission, between the occurrence of an event and the use of the processed data, such as for display or feedback and control purposes.

Near real time

Operations:

- must meet estimated time of arrival (ETA) (usually < 10 minutes)
- use predefined amount of resources
- failure needs to be fixed with a bit relaxed deadline
- have high priority

Examples:

- site monitorings
- spam detection
- speech to text in instant messaging
- video processing
  - transcoding
  - video generation
- ALS models computing
- CI/CD processes

Edge

Operations:

- must meet estimated time of arrival (ETA) (usually < 10-100 ms)
- use predefined amount of resources (stored on device)
- failure causes apps to break
- have high priority

Examples:

- IoT
- visual editors on smartphones
- bio-identification, biometry (num of steps, state of mind)
- self-driving cars
- autocomplete (T9)
- CCTV analysis

Types of compute resources

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Computation model

Types of computing power

- Bare metal
- Virtual
  - Classical VMs: KVM, vmware
  - containers: docker, podman
- On-premises or vendor cloud
  - MapReduce
  - Serverless computing (Amazon Lambda)
  - Task queues (slurm, clearml)
  - Kubernetes, k8s (Kubeflow)

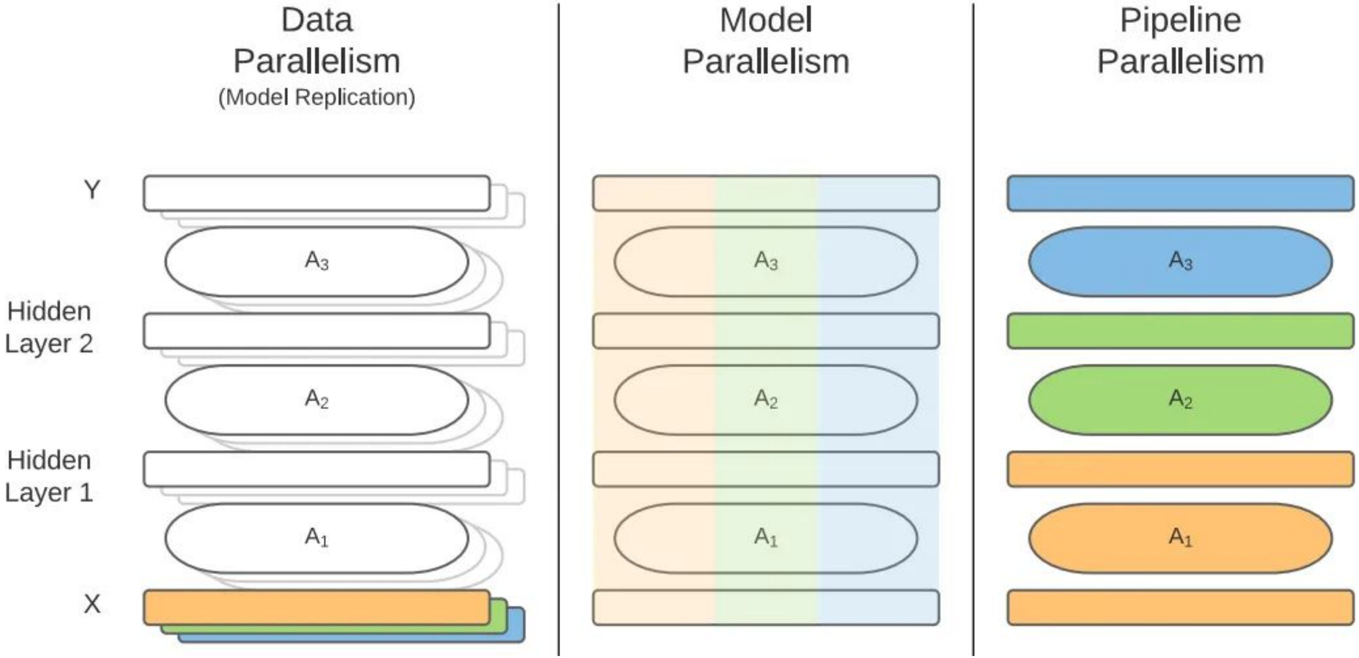
Compute speed metrics

Main metrics are

- Latency
  - how much time we wait between putting task to queue and receiving final result
- Throughput
  - how many tasks do we accomplish on average in time unit

Types of parallelization in ML

https://betterprogramming.pub/parallel-and-distributed-training-in-deep-learning-for-beginners-part-1-introduction-612a4534a117



Jobs runners

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Job runners

- MapReduce
  - Hadoop
  - YTsaurus
- Orchestration systems
  - Docker Swarm
  - Kubernetes (or k8s)
- Task queues
  - slurm
  - clearml
- Serverless
  - Amazon Lambda
- Regular job runners
  - cron
  - airflow

Regular job runners

The de facto industry standard is Airflow

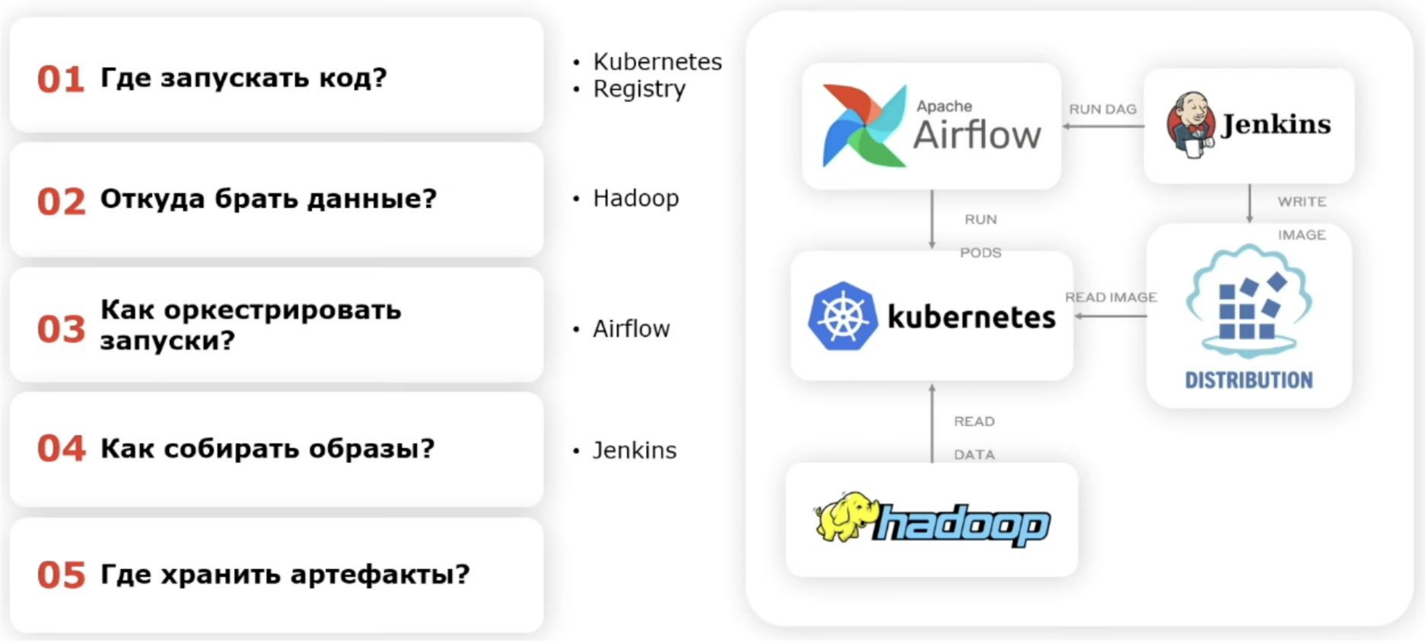
Other stacks (MapReduce, SQL DBs) have their own tools

The basic unit is Directed Acyclic Graph (DAG)

Apache Airflow

Example of autoML system

https://www.youtube.com/watch?v=HgrF4Ta2EY0



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Thanks for attention!

Questions?

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